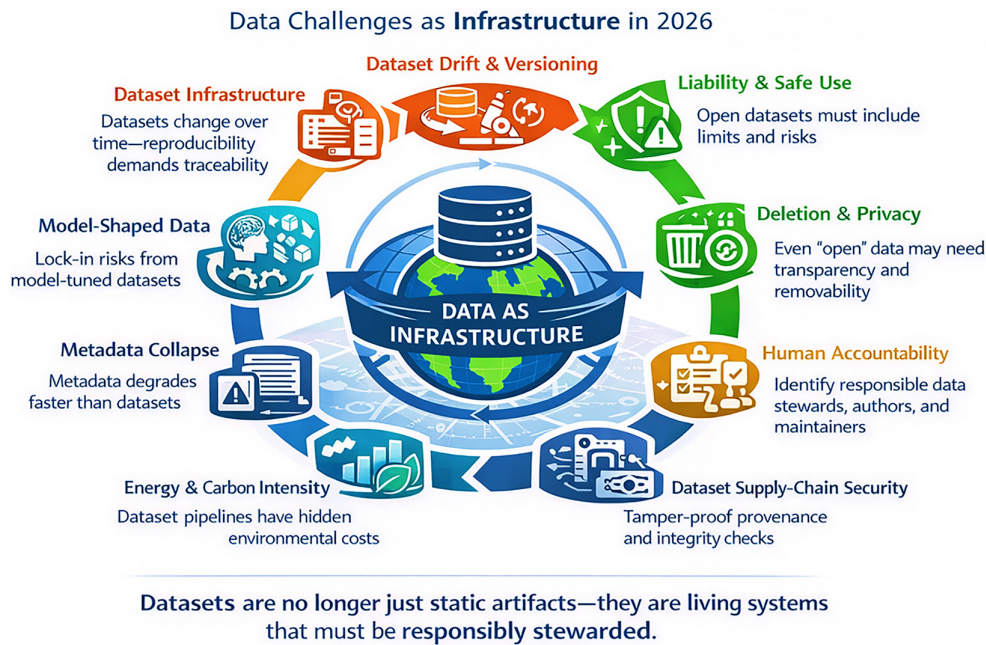


Editorial: *Data as Infrastructure*



INTRODUCTION

For more than a decade, the open data movement has advanced a compelling and necessary goal: expanding access to data in support of reproducible research, innovation, and public benefit. That goal has largely been achieved. Across disciplines, datasets are now routinely shared, cited, and reused, often at scales unimaginable when open data principles were first articulated. Yet success has introduced a new and less visible challenge. As datasets become deeply embedded in artificial intelligence systems, public infrastructure, and policy decision-making, access alone is no longer sufficient [1], [2]. In 2026, the most consequential risks to trustworthy science and responsible engineering will not arise from data scarcity, but from openly shared datasets whose assumptions, limitations, and long-term responsibilities are insufficiently documented.

This shift matters profoundly for a community committed to advancing technology for humanity. IEEE has long emphasized rigor, transparency, safety, and service to the public interest as foundational values. These principles must now be applied not only to algorithms, hardware, and systems, but to data itself [3]. Datasets increasingly function as enduring technical artifacts, shaping models, influencing decisions, and persisting far beyond their original research

context. Treating them as static supplements to publications understates both their power and their risk. The next phase of open data requires a transition from openness defined by availability to openness defined by accountability [2], [4].

NOT JUST ACCESS BUT INFRASTRUCTURE

Historically, datasets were published as supporting materials: snapshots collected for a specific study, released to enable validation or reuse. Today, many datasets operate more like infrastructure [2]. They are continuously accessed, mirrored, updated, and embedded in downstream systems. They inform energy planning, transportation policy, healthcare analytics, and autonomous decision-making. In this role, datasets resemble power grids, communication networks, or software platforms more than appendices to papers.

Infrastructure carries expectations. It must be maintained, monitored, versioned, and governed [4]. Its failure modes must be understood and communicated. Yet dataset documentation practices have not fully evolved to reflect this reality. Many datasets are released with minimal discussion of how they may change over time, how errors will be corrected, or how users should interpret updates. As a result, researchers and practitioners often assume stability where none exists. When results cannot be reproduced, the cause

is increasingly not the model or method, but undocumented changes in the underlying data.

Recognizing datasets as infrastructure re-frames documentation as an engineering responsibility rather than an administrative task. It also foregrounds stewardship: datasets, like infrastructure, require care across their lifecycle, not just at the moment of publication [5].

WHERE PRACTICES QUIETLY FALL SHORT

Several emerging issues illustrate the widening gap between the impact of datasets and their documentation. One such issue is silent dataset drift. Many datasets evolve through cleaning, augmentation, reprocessing, or partial replacement of records [6]. These changes are often well-intentioned, but they are rarely documented with the rigor applied to software versioning. Citations typically refer to a dataset by name rather than by state. As a result, different users may unknowingly work with materially different versions of *the same* dataset, undermining reproducibility and comparability.

Another under-discussed trend is the rise of model-shaped data. Increasingly, datasets are curated to align with the constraints of specific machine learning architectures, tokenization schemes, context windows, embedding pipelines, or benchmark conventions [7]. While this can improve short-term performance, it risks embedding model assumptions into the data itself. Over time, such practices create lock-in at the data layer, limiting reuse by future models and distorting the representation of real-world phenomena.

Metadata, long recognized as essential to FAIR data principles [2], faces its own scaling challenges. As datasets grow larger and more complex, metadata is often auto-generated, copied forward, or inherited without validation. Provenance becomes opaque, inconsistencies multiply, and errors propagate. The result is metadata that appears comprehensive but lacks integrity. Without mechanisms to observe, audit, and maintain metadata quality over time, even well-intentioned documentation can degrade [4].

ACCOUNTABILITY GAPS AND SOCIETAL IMPACT

Beyond technical concerns, datasets now raise questions of accountability that extend into ethics, safety, and governance. Open licenses typically disclaim liability, but disclaimers do not eliminate responsibility when data influences real-world outcomes. Datasets reused beyond their original scope can amplify bias, misinform policy, or contribute to unsafe system behavior. Dataset descriptions must explicitly articulate intended use, prohibited use, and known failure modes.

Legal and ethical pressures further complicate matters. Privacy regulations increasingly recognize rights to erasure and correction, while open datasets are often replicated across archives and mirrors. Once a dataset is cited and reused, removing or correcting harmful records becomes technically and socially challenging. The absence of shared norms for dataset deprecation, redaction propagation, or

ethical end-of-life management leaves both data producers and users navigating uncertainty.

Trust and security represent another emerging frontier. As datasets gain prominence, they become targets for manipulation. Dataset poisoning, benchmark gaming, and subtle integrity attacks threaten entire research domains. While significant attention has been paid to model security, datasets themselves remain comparatively unprotected. Treating datasets as critical infrastructure requires provenance tracking, integrity verification, and tamper-proofing which are underdeveloped in open data publishing [4].

ENERGY AND SUSTAINABILITY HIDDEN COSTS

Discussions of sustainability in computing have primarily focused on model training and inference. Yet datasets themselves carry substantial and often unmeasured environmental costs. Data collection, sensor operation, cleaning pipelines, storage, replication, and long-term retention all consume energy and resources. Large datasets may be mirrored across continents, refreshed on fixed schedules, or stored indefinitely without regard to marginal benefits.

As energy systems become more constrained and sustainability expectations rise, these costs will become harder to ignore. Funding agencies, infrastructure operators, and policymakers are beginning to ask not only whether data is valuable, but whether it is responsibly maintained. Incorporating energy and carbon considerations into dataset documentation, while still rare, aligns closely with IEEE's commitment to sustainable technology and public welfare.

OUR PUBLICATION'S ROLE

In an evolving AI landscape, *IEEE Data Descriptions* occupies a distinctive and essential role in IEEE and beyond. Our journal is not merely a repository for datasets, but a venue where expectations for responsible data publishing are articulated and refined. By emphasizing rigorous documentation, transparency, and stewardship, our journal supports reproducibility, enables trust in data-driven systems, and advances IEEE's broader mission. Importantly, this role calls for leadership that encourages authors to think critically about the dataset lifecycle, guiding reviewers to evaluate accountability alongside accessibility, and supporting infrastructure that enables long-term data stewardship [6].

Automation has brought efficiency to data management, but it also risks displacing human judgment from documentation. Domain experts possess tacit knowledge about data limitations, edge cases, and contextual nuances that automated tools cannot capture. When metadata is generated without human oversight, critical warnings and interpretations may be lost. *IEEE Data Descriptions* seeks to promote human stewardship as a complement to automation. Authors of published articles are recognized for their scholarly contributions, which reinforces accountability and professionalism. Such practices echo long-standing norms in engineering,

where responsibility is clearly assigned, and expertise is valued.

A CALL TO ACTION

Meeting the challenges of data as infrastructure requires coordinated action. Authors are encouraged to treat dataset documentation as an engineering responsibility, clearly articulating assumptions, limitations, intended use, and expectations for maintenance and change. Reviewers are invited to look beyond completeness and availability and assess whether the documentation supports reproducibility, transparency, and responsible reuse. Data stewards and infrastructure providers must provide mechanisms for richer metadata, version transparency, and long-term accountability.

IEEE Data Descriptions invites the community to lead this transition. By raising expectations thoughtfully and collaboratively raising expectations, the journal seeks to strengthen trust in open data and reinforce its role as a foundation for responsible innovation. In 2026 and beyond, the central question is no longer whether data should be open, but whether it is documented with the rigor, foresight, and stewardship required to serve humanity safely, sustainably, and reproducibly.

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